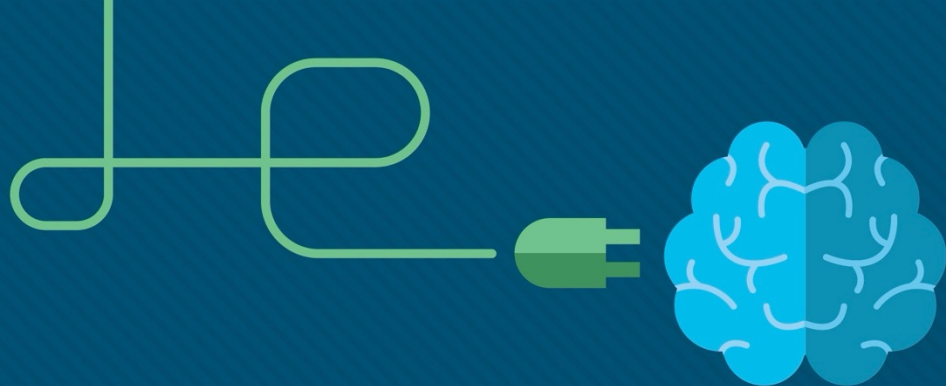




Chapter 3: Advanced STP Tuning

Instructor Materials

CCNP Enterprise: Core Networking



Chapter 3 Content

This chapter covers the following content:

- **STP Topology Tuning** - This section explains some of the options for modifying the root bridge location or moving blocking ports to designated ports.
- **Additional STP Protection Mechanisms** - This section examines protection mechanisms such as root guard, BPDU guard, and STP loop guard.

STP Topology Tuning

- In a properly designed network a switch is deliberately selected to become the root bridge and the designated and alternate ports are modified.
- Network design considerations factor in hardware platform, resiliency, and network topology.

STP Topology Tuning

Root Bridge Placement

To ensure root bridge placement set the system priority on:

- The root bridge to the lowest value
- The secondary root bridge to a value slightly higher than that of the root bridge
- All other switches to a value higher than the secondary root bridge

Command	Description
spanning-tree vlan <i>vlan-id</i> priority <i>priority</i>	The priority is a value between 0 and 61,440, in increments of 4,096.
spanning-tree vlan <i>vlan-id</i> root {primary secondary} [diameter <i>diameter</i>]	The primary keyword sets the priority to 24,576, and the secondary keyword sets the priority to 28,672. The optional diameter command makes it possible to tune the Spanning Tree Protocol (STP) convergence and modifies the timers.

STP Topology Tuning

Configuring the Root Bridge

In the example:

- The initial priority for VLAN 1 on SW1 is verified, 32,769.
- SW1 is configured to be the primary root for VLAN 1
- The priority is verified again to ensure the change took place.

Example 3-1 Changing the STP System Priority on SW1

! Verification of SW Priority before modifying the priority

SW1# show spanning-tree vlan 1

VLAN001

Spanning tree enabled protocol rstp

Root ID Priority 32769
 Address 0062.ec9d.c500
 This bridge is the root
 Hello Time 2 sec Mx Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0062.ec9d.c500
 Hello Time 2 sec Mx Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

! Configuring the SW priority as primary root for VLAN 1

SW1(config)# spanning-tree vlan 1 root primary

! Verification of SW Priority after modifying the priority

SW1# show spanning-tree vlan 1

VLAN001

Spanning tree enabled protocol rstp

Root ID Priority 24577
 Address 0062.ec9d.c500
 This bridge is the root
 Hello Time 2 sec Mx Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
 Address 0062.ec9d.c500
 Hello Time 2 sec Mx Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi 1/0/2	Desg	FWD	4	128.	2	P2p
Gi 1/0/3	Desg	FWD	4	128.	3	P2p
Gi 1/0/14	Desg	FWD	4	128.	14	P2p

STP Topology Tuning

Configuring the Backup Root Bridge

In the example:

- The initial priority for VLAN 1 on SW2 is verified, 32,769.
- SW2 is configured to be the secondary root for VLAN 1
- The priority is verified again to ensure the change took place.

Example 3-2 Changing the STP System Priority on SW2

```
! Verification of SW2 Priority before modifying the priority
SW2# show spanning-tree vlan 1
! Output omitted for brevity

VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    24577
              Address      0062.ec9d.c500
              Cost         4
              Port         1 (GigabitEthernet1/0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address      0081.c4ff.8b00
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gil/0/1        Root FWD 4         128.1    P2p
Gil/0/3        Desg FWD 4         128.3    P2p
Gil/0/4        Desg FWD 4         128.4    P2p

! Configuring the SW2 priority as root secondary for VLAN 1
SW2(config)# spanning-tree vlan 1 root secondary
```

STP Topology Tuning

Modifying STP Root Port & Blocked Switch Port Locations

Calculating total path cost to the root bridge:

- SW1 sends a BPDU to SW3 with the path cost of 0.
- SW3 receives the BPDU and adds its root port cost (4) to cost from the BPDU (0), resulting in the cost of 4.
- SW3 sends a BPDU to SW5 with the path cost of 4.
- SW5 receives the BPDU and adds its root port cost (4) to the cost from the BPDU (4), resulting in the cost of 8 for SW5 to reach the root bridge.

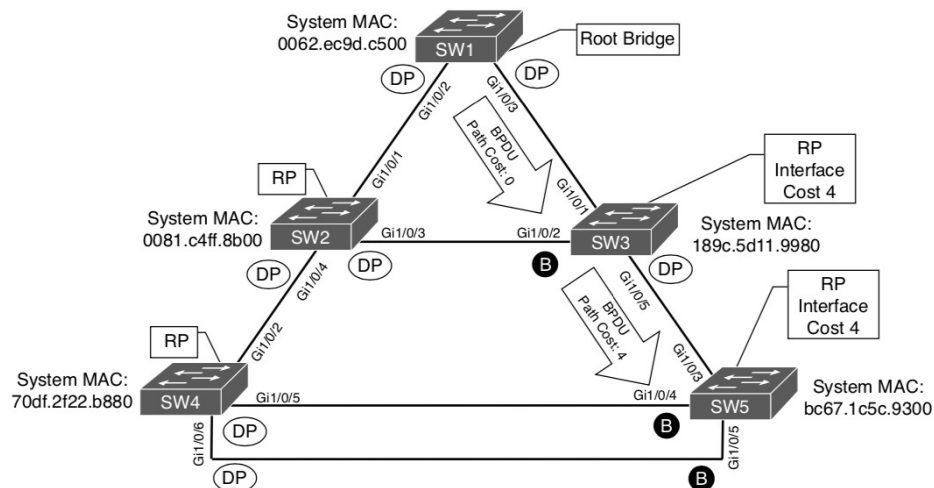


Figure 3-2 STP Path Cost Calculation

STP Topology Tuning

Verifying the Total Path Cost

The example highlights the total path cost to the root bridge from SW3 and SW5.

Example 3-3 Verifying the Total Path Cost

```
SW# show spanning-tree vlan 1
```

```
! Output omitted for brevity
```

```
VLAN0001
```

```
Root ID    Priority    32769
Address    0062.ec9d.c500
This bridge is the root
```

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi1/0/2	Desg	FWD	4	128	2	P2p
Gi1/0/3	Desg	FWD	4	128	3	P2p

```
SW# show spanning-tree vlan 1
```

```
! Output omitted for brevity
```

```
VLAN0001
```

```
Root ID    Priority    32769
Address    0062.ec9d.c500
Cost       4
Port       1 (GigabitEthernet1/0/1)
```

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi1/0/1	Root	FWD	4	128	1	P2p
Gi1/0/2	Altn	BLK	4	128	2	P2p
Gi1/0/5	Desg	FWD	4	128	5	P2p

```
SW# show spanning-tree vlan 1
```

```
! Output omitted for brevity
```

```
VLAN0001
```

```
Root ID    Priority    32769
Address    0062.ec9d.c500
Cost       8
Port       3 (GigabitEthernet1/0/3)
```

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi1/0/3	Root	FWD	4	128	3	P2p
Gi1/0/4	Altn	BLK	4	128	4	P2p
Gi1/0/5	Altn	BLK	4	128	5	P2p

Note: There is not a total path cost in SW1's output

STP Topology Tuning

Modifying STP Port Cost

- The **spanning tree [vlan vlan-id] cost cost** command can be used to modify the STP forwarding path.
- Using the spanning tree command will modify the cost for all VLANs unless the optional **vlan** keyword is used.

Example 3-4 Modifying STP Port Cost

```
SW8# conf t
SW8(config)# interface gi1/0/1
SW8(config-if)# spanning-tree cost 1
```

```
SW8# show spanning-tree vlan 1
```

```
! Output omitted for brevity
```

```
VLAN0001
```

```
Root ID    Priority    32769
           Address    0062.ec9d.c500
           Cost      1
           Port      1 (GigabitEthernet1/0/1)
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    189c.5d11.9980
```

```
..
```

```
Interface          Role Sts Cost      Prio.Nbr Type
```

```
-----
Gi1/0/1            Root FWD 1          128.1   P2p
Gi1/0/2            Desg FWD 4          128.2   P2p
Gi1/0/5            Desg FWD 4          128.5   P2p
```

```
SW8# show spanning-tree vlan 1
```

```
! Output omitted for brevity
```

```
VLAN0001
```

```
Root ID    Priority    32769
           Address    0062.ec9d.c500
           Cost      4
           Port      1 (GigabitEthernet1/0/1)
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0081.c4ff.8b00
```

```
..
```

```
Interface          Role Sts Cost      Prio.Nbr Type
```

```
-----
Gi1/0/1            Root FWD 4          128.1   P2p
Gi1/0/3            Altn BLK 4          128.3   P2p
Gi1/0/4            Desg FWD 4          128.4   P2p
```

STP Topology Tuning

Modifying STP Port Priority

STP port priority influences which port becomes the alternate port when multiple links are used between switches. Use the command **spanning-tree [vlan *vlan-id*] port-priority *priority*** to change the STP port priority on a switch's interface.

Example 3-5 Viewing STP Port Priority

```
SW# show spanning-tree vlan 1
! Output omitted for brevity
VLAN001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
            Address    0062.ec9d.c500
            Cost       12
            Port       4 (GigabitEthernet1/0/4)

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    bc67.1c5c.9300
..
Interface    Role Sts Cost    Prio.Nbr Type
-----
Gi1/0/4      Root FWD 4      128.4   P2p
Gi1/0/5      Altn BLK 4      128.5   P2p
```

Example 3-6 Verifying Port Priority Impact on an STP Topology

```
SW# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW(config)# interface gi1/0/6
SW(config-if)# spanning-tree port-priority 64
```

Example 3-7 Determining the Impact of Port Priority on a Topology

```
SW# show spanning-tree vlan 1
! Output omitted for brevity
Interface    Role Sts Cost    Prio.Nbr Type
-----
Gi1/0/2      Root FWD 4      128.2   P2p
Gi1/0/5      Desg FWD 4      128.5   P2p
Gi1/0/6      Desg FWD 4      64.6    P2p
```

```
SW# show spanning-tree vlan 1
! Output omitted for brevity
Interface    Role Sts Cost    Prio.Nbr Type
-----
Gi1/0/4      Altn BLK 4      128.4   P2p
Gi1/0/5      Root FWD 4      128.5   P2p
```

Additional STP Protection Mechanisms

- A network forwarding loop occurs when there are multiple active paths between two devices. Broadcast and multicast traffic are forwarded out every switch port continuing the forwarding loop.
- The network's throughput is drastically effected as the switches are processing numerous frames. The switches CPU utilization will be high and memory space will be consumed. The switches might crash and users will likely notice the impact on the network.

Additional STP Protection Mechanisms

Common issues for Layer 2 forwarding loops:

- STP is disabled on a switch.
- A load balancer is misconfigured and sends traffic out multiple ports with the same MAC address.
- A virtual switch that bridges two physical ports.
- End users using an unmanaged switch or hub.

Additional STP Protection Mechanisms

Root Guard

Root guard is an STP feature that prevents a configured port from becoming a root port.

- It does this by placing the port in an ErrDisabled state if a superior BDPU is received on that port.
- Root guard is placed on designated ports towards other switches that should never become root bridges.
- Root guard is enabled on a port-by-port basis.

Use the interface command **spanning-tree guard root** to enable root guard.

Additional STP Protection Mechanisms

STP Portfast

STP portfast disables the topology notification notification (TCN) generation and causes access ports that come up to bypass the learning and listening states and enter the forwarding state immediately. If a BPDU is received on a portfast-enabled port, the portfast functionality is removed from that port.

Command	Description
spanning-tree portfast	Interface command to enable portfast on a specific access port
spanning-tree portfast default	Global command to enable portfast on all access ports
spanning-tree portfast disable	Disable portfast on a port
spanning-tree portfast trunk	Command used on trunk links to enable portfast *This command should only be used with ports connected to a single host.

Additional STP Protection Mechanisms

STP Portfast Examples

The following shows how to enable STP portfast globally and on a specific interface.

Example 3-9 Enabling STP Portfast on Specific Interfaces

```
SW(config)# interface gigabitEthernet 1/0/13
SW(config-if)# switchport mode access
SW(config-if)# switchport access vlan 10
SW(config-if)# spanning-tree portfast
```

```
SW# show spanning-tree vlan 10
```

! Output omitted for brevity

VLAN0010

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi 1/0/2	Desg	FWD	4	128.2		P2p
----------	------	-----	---	-------	--	-----

Gi 1/0/3	Desg	FWD	4	128.3		P2p
----------	------	-----	---	-------	--	-----

Gi 1/0/13	Desg	FWD	4	128.13		P2p Edge
-----------	------	-----	---	--------	--	----------

```
SW# show spanning-tree interface gi1/0/13 detail
```

Port 13 (GigabitEthernet1/0/13) of VLAN0010 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.7.

Designated root has priority 32778, address 0062.ec9d.c500

Designated bridge has priority 32778, address 0062.ec9d.c500

Designated port id is 128.7, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

The port is in the portfast mode

Link type is point-to-point by default

BPDUs: sent 23103, received 0

Example 3-10 Enabling STP Portfast Globally

```
SW# conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
SW(config)# spanning-tree portfast default
```

Warning: this command enables portfast by default on all interfaces. You should now disable portfast explicitly on switched ports leading to hubs, switches and bridges as they may create temporary bridging loops.

```
SW(config)# interface gi1/0/8
```

```
SW(config-if)# spanning-tree portfast disable
```

Additional STP Protection Mechanisms

BPDU Guard

BPDU guard is a safety mechanism that shuts down ports configured with STP portfast upon receiving a BPDU.

Command	Description
spanning-tree portfast bpduguard default	Global command to enable BPDU guard on all STP portfast ports
spanning-tree bpduguard {enable disable}	Interface command to enables or disable BPDU guard on a specific interface
show spanning-tree interface <i>interface-id</i> detail	Displays whether BPDU guard is enabled for the specified interface

Note: BPDU Guard is typically configured with all host-facing ports that are enabled with portfast.

Additional STP Protection Mechanisms

BPDUGuard Examples

The following shows how to configure BPDUGuard and a BPDUGuard-enabled port detecting a BPDUGuard.

Example 3-11 *Configuring BPDUGuard*

```
SW# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW(config)# spanning-tree portfast bpduguard default
SW(config)# interface gi1/0/8
SW(config-if)# spanning-tree bpduguard disable

SW# show spanning-tree interface gi1/0/7 detail
Port 7 (GigabitEthernet1/0/7) of VLAN010 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.7.
  Designated root has priority 32778, address 0062.ec9d.c500
  Designated bridge has priority 32778, address 0062.ec9d.c500
  Designated port id is 128.7, designated path cost 0
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast mode
  Link type is point-to-point by default
  Bpd guard is enabled by default
  BPDU: sent 23386, received 0

SW# show spanning-tree interface gi1/0/8 detail
Port 8 (GigabitEthernet1/0/8) of VLAN010 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.8.
  Designated root has priority 32778, address 0062.ec9d.c500
  Designated bridge has priority 32778, address 0062.ec9d.c500
  Designated port id is 128.8, designated path cost 0
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast mode by default
  Link type is point-to-point by default
  BPDU: sent 23388, received 0
```

Example 3-12 *Detecting a BPDUGuard on a BPDUGuard-Enabled Port*

```
12: 47: 02. 069: %SPANTREE-2- BLOCK_BPDUGUARD: Received BPDUGuard on port Gigabit
Ethernet1/0/2 with BPDUGuard enabled. Disabling port.
12: 47: 02. 076: %PM-4- ERR_DISABLE: bpduguard error detected on Gi1/0/2,
putting Gi1/0/2 in err-disable state
12: 47: 03. 079: %LINKPROTO-5-UPDOWN: Line protocol on Interface Gigabit
Ethernet1/0/2, changed state to down
12: 47: 04. 082: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/2, changed
state to down
```

```
SW# show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi1/0/1		not connect	1	auto	auto 10/100/1000	BaseTX
Gi1/0/2	SW Gi1/0/1	err-disabled	1	auto	auto 10/100/1000	BaseTX
Gi1/0/3	SW Gi1/0/1	connected	trunk	auto	10/100/1000	BaseTX

The **show interfaces status** command shows the err-disabled status of the port that received the BPDUGuard.

Additional STP Protection Mechanisms

BPDU Guard Error Recovery

The Error Recovery service can be used to reactivate ports that are shut down. Ports that are put into the ErrDisabled mode due to BPDU guard do not automatically restore themselves. Use the following commands to recover ports that were shutdown from BPDU guard:

Command	Description
errdisable recovery cause bpduguard	Recovers ports shutdown by BPDU guard
errdisable recovery interval <i>time-seconds</i>	The period that Error Recovery checks for ports

Additional STP Protection Mechanisms

BPDUGuard Error Recovery Example

The following example shows how to configure the Error Recovery service.

Example 3-13 Configuring Error Recovery Service

```
SW# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW(config)# errdisable recovery cause bpduguard
```

```
SW# show errdisable recovery
! Output omitted for brevity
ErrDisable Reason    Timer Status
-----
arp-inspection        Disabled
bpduguard             Enabled
..
Recovery command: "clear" Disabled
Timer interval: 300 seconds
```

Interfaces that will be enabled at the next timeout:

Interface	Errdisable reason	Time left(sec)
-----	-----	-----
Gi1/0/2	bpduguard	295

! Syslog output from BPDUGuard recovery. The port will be recovered, and then
! triggered again because the port is still receiving BPDUs.

SW#

01:02:08.122: %PM 4-ERR_RECOVER: Attempting to recover from bpduguard err-disable
state on Gi1/0/2

01:02:10.699: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gigabit
Ethernet1/0/2 with BPDUGuard enabled. Disabling port.

01:02:10.699: %PM 4-ERR_DISABLE: bpduguard error detected on Gi1/0/2, putting
Gi1/0/2 in err-disable state

Note: The Error Recovery service operates every 300 seconds (5 minutes). This can be changed from 5 to 86,000 seconds with the global command **errdisable recovery interval time**

Additional STP Protection Mechanisms

BPDU Filter

BPDU filter blocks BPDUs from being transmitted out of a port. It can be enabled globally or on a specific interface.

Global BPDU filter command:

spanning-tree portfast bpdupfilter default

With the global BPDU configuration the port sends a series of 10 – 12 BPDUs. If the switch receives any BPDUs, it checks to identify which switch is more preferred.

- The preferred switch doesn't process any BPDUs but still passes them along to inferior switches.
- A non-preferred switch processes the BPDUs that are received but doesn't transmit any BPDUs to superior switches.

Interface-specific BPDU filter command:

Spanning-tree bpdupfilter enable

With the interface-specific BPDU configuration the port does not send any BPDUs on an ongoing basis. If the remote port has BPDU guard, that generally shuts down the port as a loop prevention mechanism.

Additional STP Protection Mechanisms

Verifying a BPDU Filter

The following shows using the **show spanning-tree interface *interface-id* detail** command to verify that BPDU filter is enabled.

Example 3-14 Verifying a BPDU Filter

```
! SW was enabled with BPDU filter only on port Gi1/0/2
SW# show spanning-tree interface gi1/0/2 detail | in BPDU Bpdu Ethernet
Port 2 (GigabitEthernet1/0/2) of VLAN0001 is designated forwarding
  Bpdu filter is enabled
  BPDU: sent 113, received 84
SW# show spanning-tree interface gi1/0/2 detail | in BPDU Bpdu Ethernet
Port 2 (GigabitEthernet1/0/2) of VLAN0001 is designated forwarding
  Bpdu filter is enabled
  BPDU: sent 113, received 84
```

```
! SW was enabled with BPDU filter globally
SW# show spanning-tree interface gi1/0/2 detail | in BPDU Bpdu Ethernet
Port 1 (GigabitEthernet1/0/2) of VLAN0001 is designated forwarding
  BPDU: sent 56, received 5
SW# show spanning-tree interface gi1/0/2 detail | in BPDU Bpdu Ethernet
Port 1 (GigabitEthernet1/0/2) of VLAN0001 is designated forwarding
  BPDU: sent 58, received 5
```

Problems with Unidirectional Links

Network devices that utilize fiber-optic cables for connectivity can encounter unidirectional traffic flows if one strand is broken. BPDUs will not be able to be transmitted causing other switches on the network to eventually time out the existing root port and change root ports resulting in a forwarding loop.

Two solutions to problems with unidirectional links:

- STP Guard
- Unidirectional Link Detection

Additional STP Protection Mechanisms

STP Loop Guard

STP Loop guard prevents any alternative or root ports from becoming designated ports due to loss of BPDUs on the root port. Loop guard places the original port into an ErrDisabled state while BPDUs are not being received and transitions back through the STP states when it begins receiving BPDUs again.

Command	Description
spanning-tree loopguard default	Global command to enable loop guard
spanning-tree guard loop	Interface command to enable loop guard
show spanning-tree inconsistent-ports	Shows ports in the inconsistent state due to the port not receiving BPDUs

Note: Loop guard shouldn't be enabled on portfast-enabled ports because it directly conflicts with root/alternate port logic

Additional STP Protection Mechanisms

STP Loop Guard Examples

The following examples show configuring loop guard, triggering loop guard by blocking BPDUs and the port in an inconsistent state.

Example 3-15 Configuring Loop Guard

```
SW2# config t
SW2(config)# interface gi1/0/1
SW2(config-if)# spanning-tree guard loop
! Placing BPDU filter on SW2's RP (Gi1/0/1) bridge) triggers loop guard.
SW2(config-if)# interface gi1/0/1
SW2(config-if)# spanning-tree bpduguard enabled
01:42:35.051: %SPANTREE-2-LOOPGUARD_BLOCK: Loop guard blocking port Giabit
Ethernet1/0/1 on VLAN001
```

```
SW2# show spanning-tree vlan 1 | b Interface
Interface          Role Sts Cost      Prio.Nbr Type
-----
Gi 1/0/1           Root BKN*4      128.1   P2p *LOOP_Inc
Gi 1/0/3           Root FWD 4        128.3   P2p
Gi 1/0/4           Desg FWD 4        128.4   P2p
```

Example 3-16 Viewing the Inconsistent STP Ports

```
SW2# show spanning-tree inconsistent ports
```

Name	Interface	Inconsistency
VLAN001	Gi gabitEthernet1/0/1	Loop Inconsistent
VLAN010	Gi gabitEthernet1/0/1	Loop Inconsistent
VLAN020	Gi gabitEthernet1/0/1	Loop Inconsistent
VLAN099	Gi gabitEthernet1/0/1	Loop Inconsistent

```
Number of inconsistent ports (segments) in the system: 4
```


Additional STP Protection Mechanisms

Unidirectional Link Detection

Unidirectional Link Detection (UDLD) allows for the bidirectional monitoring of fiber-optic cables.

UDLD operates in two modes:

- **Normal** – If a frame is not acknowledged, the link is considered undetermined and the port remains active.
- **Aggressive** – If a frame is not acknowledged, the switch sends another 8 packets in 1 second intervals. If those packets aren't acknowledged, the port is placed into an error state.

Additional STP Protection Mechanisms

UDLD Commands

The following are commands for configuring and verifying UDLD:

Command	Description
udld enable [aggressive]	Global command to enable UDLD. *Optional aggressive keyword sets the mode to aggressive.
udld port [aggressive]	Interface command to enable UDLD *Optional aggressive keyword sets the mode to aggressive.
udld port disable	Disable UDLD on a specific interface
udld recovery [interval <i>time</i>]	Enables UDLD recovery. The <i>time</i> default value is 5 minutes.
show udld neighbors	Displays the status of UDLD neighborship
show udld <i>interface-id</i>	Displays detailed information about UDLD

Additional STP Protection Mechanisms

Configuring & Verifying UDLD Examples

The following are examples for configuring and verifying UDLD:

Example 3-17 Configuring UDLD

```
SW# conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW(config)# udld enable
```

Example 3-18 Verifying UDLD Neighbors and Switch Port Status

```
SW# show udld neighbors

Current operational state: Advertisement - Single neighbor detected
Message interval: 15000 ms
Time out interval: 5000 ms

Port fast-hello configuration setting: Disabled
Port fast-hello interval: 0 ms
Port fast-hello operational state: Disabled
Neighbor fast-hello configuration setting: Disabled
Neighbor fast-hello interval: Unknown

Entry 1
---
Expiration time: 41300 ms
Cache Device index: 1
Current neighbor state: Bidirectional
Device ID: 081C4FF8B0
Port ID: Te1/1/3
Neighbor echo 1 device: 062EC9DC50
Neighbor echo 1 port: Te1/1/3

TLV Message interval: 15 sec
No TLV fast-hello interval
TLV Time out interval: 5
TLV CDP Device name: SW2
```

Prepare for the Exam

Prepare for the Exam

Key Topics for Chapter 3

Description
Root bridge placement
Root bridge values
Spanning tree port cost
Root guard
STP portfast
BPDU guard
BPDU filter

Prepare for the Exam

Key Terms for Chapter 3

Terms
BPDU filter
Root guard
STP loop guard
BPDU guard
STP portfast
Unidirectional Link Detection (UDLD)

Command Reference for Chapter 3

Task	Command Syntax
Configure the STP priority for a switch so that it is a root bridge or a backup root bridge	spanning-tree vlan vlan-id root { primary secondary } [diameter diameter] OR spanning-tree vlan vlan-id priority priority
Configure the STP port cost	spanning tree [vlan vlan-id] cost cost
Configure the STP port priority on the downstream port	spanning-tree [vlan vlan-id] port-priority priority
Enable root guard on an interface	spanning-tree guard root
Enable STP portfast globally, for a specific port, or for a trunk port	spanning-tree portfast default OR spanning-tree portfast OR spanning-tree portfast trunk
Enable BPDU guard globally or for a specific switch port	spanning-tree portfast bpduguard default OR spanning-tree bpduguard { enable disable }

Command Reference for Chapter 3 (Cont.)

Task	Command Syntax
Enable BPDU guard globally or for a specific interface	spanning-tree portfast bpdupfilter default OR spanning-tree bpdupfilter enable
Enable STP loop guard globally or for a specific interface	spanning-tree loopguard default OR spanning-tree guard loop
Enable automatic error recovery for BPDU guard.	errdisable recovery cause bpduguard
Enable BPDU guard globally or for a specific interface	spanning-tree portfast bpdupfilter default OR spanning-tree bpdupfilter enable
Enable STP loop guard globally or for a specific interface	spanning-tree loopguard default OR spanning-tree guard loop
Enable automatic error recovery for BPDU guard.	errdisable recovery cause bpduguard

Command Reference for Chapter 3 (Cont.)

Task	Command Syntax
Change the automatic error recovery time	errdisable recovery interval time-seconds
Enable UDLD globally or for a specific port	udld enable [aggressive] OR udld port [aggressive]
Display the list of STP ports in an inconsistent state	show spanning-tree inconsistentports
Display the list of neighbor devices running UDLD	show udld neighbors

